Host and burst properties vs axis ratio: outlier-robust statistics

Property vs axis ratio q	N	r_S (p)	bootstrap 95%	Kendall τ (p)	delete-1 range	Theil–Sen slope [95%]
DM_{host} (rest frame)	55	-0.06 (0.64)	[-0.32, +0.20]	-0.04 (0.63)	-0.12 to -0.04	-38 [-239, 159]
DM_{host}, foreground halos removed	49	-0.04 (0.76)	[-0.33, +0.25]	-0.03 (0.74)	-0.10 to +0.00	-51 [-288, 165]
DM_{host}, foreground excess over random sightlines removed	49	-0.08 (0.60)	[-0.35, +0.21]	-0.06 (0.56)	-0.13 to -0.03	-67.6 [-278, 147]
DM_{host}, foreground removed (all photo-z galaxies in front)	49	-0.07 (0.62)	[-0.35, +0.22]	-0.04 (0.67)	-0.12 to -0.02	-42.6 [-292, 178]
DM_{host}, spectroscopic foreground only	49	-0.06 (0.68)	[-0.34, +0.23]	-0.04 (0.72)	-0.12 to -0.02	-32.6 [-243, 147]
log|RM|	41	-0.02 (0.91)	[-0.31, +0.30]	-0.00 (0.99)	-0.09 to +0.04	-0.00544 [-1.46, 1.12]
$\log \tau_{scatt}$	29	-0.17 (0.39)	[-0.55, +0.27]	-0.11 (0.41)	-0.28 to -0.07	-1.03 [-2.92, 1.48]
Linear polarization	24	-0.33 (0.11)	[-0.69, +0.12]	-0.24 (0.096)	-0.45 to -0.25	-0.467 [-1.08, 0.188]
A_V, old stars	30	-0.46 (0.010)	[-0.67, -0.15]	-0.28 (0.032)	-0.53 to -0.41	-0.429 [-1.06, -0.0207]
DM_{host} (rest frame), m_r < 21	45	-0.09 (0.54)	[-0.38, +0.19]	-0.07 (0.52)	-0.16 to -0.06	-54 [-273, 168]
DM_{host} (rest frame), n < 2	46	-0.04 (0.81)	[-0.32, +0.26]	-0.02 (0.85)	-0.10 to +0.00	-10.5 [-177, 168]
DM_{host} (rest frame), bulge fits removed	52	-0.07 (0.64)	[-0.34, +0.20]	-0.04 (0.64)	-0.12 to -0.04	-38.4 [-254, 168]
DM_{host}, foreground halos removed, m_r < 21	39	-0.04 (0.81)	[-0.36, +0.28]	-0.03 (0.76)	-0.11 to +0.02	-41.5 [-256, 173]
DM_{host}, foreground halos removed, n < 2	42	-0.00 (0.99)	[-0.31, +0.30]	-0.00 (0.99)	-0.06 to +0.05	-0.495 [-274, 204]
DM_{host}, foreground halos removed, bulge fits removed	46	-0.04 (0.79)	[-0.33, +0.26]	-0.03 (0.77)	-0.10 to +0.01	-46.9 [-290, 167]

Bootstrap: 10,000 resamples. Since every host has $q > q_0$, rank correlations with q and with $\cos i$ are identical (sign flipped for i).
A_V of the old stars is the only $p < 0.05$ case (dust in inclined disks); it is not significant after correcting for the 41 properties tested.